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S-adenosyl-L-methionine is the unexpected methyl donor for the methylation of mercury by the membrane-associated HgcAB complex

Kaiyuan Zheng[a,1], Katherine W. Rush[a,2], Swapneeta S. Date[b,3], Alexander Johs[b,3], Jerry M. Parks[c], Angela S. Fleischhacker[a], Macon J. Abernathy[d], Ritimukta Sarangi[d], and Stephen W. Ragsdale[a,4]

Affiliations are included on p. 10.

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Mercury (Hg) is a heavy metal that exhibits high biological toxicity. Monomethylmercury and dimethylmercury are neurotoxins and a significant environmental concern as they bioaccumulate and biomagnify within the aquatic food web. Microbial Hg methylation involves two proteins, HgcA and HgcB. Here, we show that HgcA and HgcB can be heterologously coexpressed, and the HgcAB complex can be purified. We demonstrated that HgcA is a membrane-associated cobalamin-dependent methyltransferase and HgcB is a ferredoxin-like protein containing two [4Fe-4S] clusters. Further, spectroscopic and kinetic results demonstrate that S-adenosyl-L-methionine (SAM) donates the methyl group to Hg in a two-step reaction involving a methylcob(III)alamin intermediate including Co-thiolate ligation from a conserved Cys residue. Our findings uncover a biological role for SAM in microbial Hg methylation.

methyl mercury | S-adenosyl-L-methionine | cobalamin | iron-sulfur | methyltransferase

Significance

The microbial conversion of inorganic mercury to highly neurotoxic methylmercury is a global environmental concern and a threat to human health. Most mercury methylation research is focused on genomic and microbial aspects, and biochemical studies have been challenging because of the difficulties in expressing and purifying the proteins involved in mercury methylation (HgcAB). The present study establishes the active expression, purification, and characterization of HgcAB as a B_{12}/iron–sulfur enzyme that catalyzes S-adenosyl methionine (SAM)-dependent Hg methylation through a methyl-Co-thiolate intermediate. This work identifies a metabolic role for SAM in heavy metal–associated biological processes.

Mercury (Hg) is a pervasive global pollutant and health threat that predominantly occurs in the environment as elemental Hg^0, Hg^{++}, and its highly toxic forms, methylmercury (CH_3Hg^+, MeHg) and dimethylmercury [$Hg(CH_3)_2$, Me_2Hg] (1). Hg is released into ecosystems from natural geothermal activity as well as from anthropogenic sources, including mining operations, coal combustion, and chemical industries (2). Microorganisms convert Hg^{++} into MeHg, which bioaccumulates and biomagnifies up trophic levels throughout the food chain (3–5) and is responsible for Minamata disease (6). Understanding the fundamental mechanism of MeHg formation will promote the development of effective strategies to combat exposure to this neurotoxin and may provide insights into how other heavy metal(loid) (i.e., As, Sn) homeostasis pathways produce harmful methylated metal(loid) products.

Work in the early 1990s indicated that MeHg formation is a cobalamin-dependent enzymatic process (7). More recently, the two genes, *hgcA* and *hgcB*, involved in mercury methylation were identified (3) and subsequent phylogenic analyses identified mercury-methylating microorganisms within many phyla, including sulfate- and iron-reducing bacteria, Deltaproteobacteria, Firmicutes, Geobacterales, Nitrospirae, Spirochaetes, Actinobacteria, and Euryarchaeota (8–10), but with no obvious metabolic trends (11–13). Bioinformatic analyses and structural modeling (14) indicate that HgcA consists of an N-terminal cobalamin binding domain and a C-terminal transmembrane domain consisting of five transmembrane α-helices (3), while HgcB contains two CxxCxxCxxxCP motifs proposed to bind [4Fe-4S] clusters (14–17). Mutagenesis studies identified a conserved cysteine residue (C93) in HgcA (18) and three cysteine residues on HgcB (C73, C94, and C95) that appear to be crucial for mercury methylation activity (3, 18). Moreover, recent sequence and codon analyses suggested that some homologs of *hgcA* and *hgcB* may be selenoproteins (19).

Methyltetrahydrofolate (Me-THF), a common methyl donor in cobalamin-dependent methyl transferases (20–22), was proposed to be the Hg methyl donor (7). DFT calculations suggested the reaction occurs through a methyl carbanion or radical intermediate (23, 24). Nevertheless, it has been challenging to mechanistically characterize the reaction(s) catalyzed by HgcAB due to its very low expression levels (~0.0004% of total protein) (25, 26), membrane association, oxygen sensitivity, and the need to incorporate cobalamin and Fe-S clusters for activity. Here, we report an expression and purification protocol for the HgcAB complex that improves yields and activity, allowing us to characterize the HgcAB complex, and demonstrate that SAM is the methyl donor.

The authors declare no competing interest.

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[1]Present Address: Department of Structural Molecular Biology, University of Virginia School of Medicine, Charlottesville, VA 22908.

[2]Present Address: Department of Chemistry and Biochemistry, Auburn University, Auburn, AL 36849-5312.

[3]Present Address: Center for Technology Transfer and Commercialization, Vanderbilt University, Nashville, TN 37212-2838.

[4]To whom correspondence may be addressed. Email: sragsdal@umich.edu.

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Results

Purification of the HgcAB Complex. While important strides have been made by microbiological, genetic, and bioinformatic studies, no successful purification protocol of the active HgcAB complex had been established. Prior mutagenesis studies indicated that the transmembrane domain of HgcA is required for Hg methylation activity (18). Furthermore, our previous efforts at independent heterologous expression of either HgcA or HgcB in *Escherichia coli* produced insoluble proteins lacking cobalamin, which indicated that we should coexpress HgcA and HgcB together via a N-terminal His-tagged full-length HgcA and a tagless HgcB. Thus, we cloned the His-tagged variant of the full-length *Pseudodesulfovibrio mercurii* sp. nov. strain ND132 (*P. mercurii*) (formerly *Desulfovibrio desulfuricans* ND132) (27), *hgcA* (His-*hgcA*) in tandem with *hgcB* into the pETDuet vector to generate pETDuet-His-*hgcA*-*hgc*B (28). To fully load HgcAB with its cobalamin center and two [4Fe-4S] clusters, we cotransformed the pETDuet-His-*hgcA*-*hgcB* vector with the pRKISC plasmid (29–31) for Fe-S cluster assembly (graciously provided by Yasuhiro Takahashi, Osaka University) and the pBtu plasmid for cobalamin uptake (32) vectors into *E. coli* C41(DE3), which often aids in expression of membrane-associated proteins, and BL21(DE3) strains (32). Both C41(DE3) and BL21(DE3) cells were cultivated under aerobic conditions to reach OD_{600} ~0.35 and induced by arabinose and then reach OD_{600} ~0.9 and induced by IPTG for the *btu*- and *isc*-operon-encoded proteins and for HgcAB overexpression under anaerobic conditions. While the C41 cell lysate did not exhibit mercury methylation activity, that from BL21 did catalyze MeHg formation (*SI Appendix*, Fig. S1).

Incorporation of an N-terminal His-tag on HgcA and inclusion of sodium *N*-dodecanoyl-*N*-methylglycinate (Sarkosyl) and Triton X-100 detergents in the buffers enabled one-step purification of the HgcAB complex using a Ni-NTA resin (33), as shown by native- and sodium dodecyl sulfate - polyacrylamide gel electrophoresis (SDS-PAGE) and tandem mass spectrometric (MS/MS) analyses (*SI Appendix*, Figs. S2 and S3*C*). After buffer exchange and dialysis to remove excess salt and detergent, HgcAB exhibited ultraviolet (UV)-visible spectra indicating high levels of cobalamin. Quantification of the cobalamin by conversion to dicyano-Co(III)balamin (34, 35) demonstrated 70% incorporation (*SI Appendix*, Fig. S3 *A* and *B*). Furthermore, electron paramagnetic resonance (EPR) spectral analysis of the dithionite-reduced protein indicated incorporation of two iron–sulfur clusters ([4Fe-4S]/Cbl ~1.72) (*SI Appendix*, Fig. S4), as predicted previously based on sequence analysis (3, 14).

We tried various detergents to stabilize and purify the active membrane-associated HgcAB complex and to perform the experiments described here. For example, SDS denatured HgcAB, while CHAPS and octylglucopyranoside caused HgcAB precipitation, preventing cobalamin incorporation. Using only Sarkosyl in purification allowed solubilization of HgcAB, enabling Ni-NTA affinity chromatography purification, but inhibited methylation of the purified protein complex. Triton X-100 alone also resolubilized and allowed purification of an HgcAB complex that is active in the first half-reaction, but the yield was extremely low compared to the Sarkosyl-Triton combination.

Demonstration that SAM Is the Methyl Donor. Prior studies have suggested that the methyl donor for mercury methylation is Me-THF (7, 12). However, enzymatic analyses of the overall mercury methylation reaction by cell lysates (Fig. 1) and enzymatic studies of the first half-reaction (cobalamin methylation) with the purified HgcAB complex (Fig. 2 *B*–*D*) do not support this hypothesis. Of the potential methyl donors tested, including Me-THF, acetyl coenzyme A (acetyl-CoA), serine, glycine, methionine, and SAM, only the inclusion of SAM in the assay led to a significant increase in MeHg formation above background levels in cell lysates (Fig. 1*A*). We also observed a positive correlation between SAM concentration and MeHg yield until the SAM concentration reached saturation around 150 μM (Fig. 1*B*). Cell lysate from BL21(DE3) expressing HgcAB exhibited increased Hg methylation activity (Fig. 1*C*) compared with the negative control (*Left* panel, rows 1 and 2). Importantly, inclusion of SAM instead of Me-THF produced significantly higher MeHg yields in *E. coli* lysates containing HgcAB (Fig. 1 *C*, *Left* panel, rows 3 vs. 4, 5 vs. 6), independent of whether HgcA contained (*Left* panel, rows 5 and 6) or lacked (*Left* panel, rows 3 and 4) an N-terminal His_6-tag. Similar results were observed with *E. coli* lysates containing HgcAB mixed with Δ*hgcAB* ND132 cell lysate (Fig. 1 *C*, *Middle* panel, rows 7 vs. 8, 9 vs. 10 regardless of the His_6-tag). Similar results were also obtained when the assays contained partially purified HgcAB instead of cell lysates (Fig. 1 *C*, *Middle* panel, rows 11 vs. 12).

The high background levels of MeHg in Fig. 1 are due to endogenous SAM in the cell lysates. Unfortunately, since SAM is such a crucial methyl donor involved in numerous cellular processes, we do not yet have a method to cleanly deplete SAM. However, even with the background level of MeHg, the addition of SAM resulted in a statistically significant increase in MeHg production. Including Me-THF (Fig. 1*A*, lane control and lane Me-THF) did not produce a statistically different yield of MeHg compared with controls. These data indicate that HgcA utilizes SAM, not Me-THF, for Hg methylation. We also can conclude that the N-terminal His_6-tag on HgcA (Fig. 1 *C*, *Left* panel, rows 3 vs. 5) does not significantly affect the level of methylation. However, the overall Hg methylation activity is inhibited by Sarkosyl (Fig. 1 *C*, *Right* panel, rows 16 vs. 17), which was used during purification, as shown in cell lysate assays. This finding explains why the purified HgcAB complex does not exhibit mercury methylation activity in Δ*hgcAB* ND132 cell lysate (Fig. 1 *C*, *Right* panel, row 13 to 15). On the other hand, as shown in experiments with the purified protein, Sarkosyl does not affect the first half-reaction from Cob(I)- to Me-Cob(III)-HgcAB (Fig. 2*C*).

To further characterize the role of SAM as the methyl donor, we explored the formation of S-adenosyl-L-homocysteine (SAH), which should be formed in stoichiometric amounts relative to those of Cob(I)-HgcAB during methyl-Co(III) formation. Thus, we used LC–MS to separate and track the conversion of SAM (m/z = 399.14) to SAH (m/z of SAH is 385.13) during this single-turnover reaction (Fig. 2*E*). Upon addition of 150 μM SAM to 50 μM Cob(I)-HgcAB, about 50 μM SAH was formed, which equals the amount of HgcAB in the reaction (Fig. 2*E*, group c). SAH was not detected in the two control groups, when SAM was omitted and when the reaction was quenched before addition of SAM (Fig. 2*E*, group a and b). To convert this single-turnover to a multiple-turnover reaction, we reacted Cob(I)-HgcAB with excess SAM, while exposing the reaction to light from a fluorescent lamp. Under these conditions, the Me-Co(III) formed from reaction with SAM underwent photolysis to Co(II) followed by Ti(III)-dependent reduction to Cob(I), which reacted with the remaining SAM. Thus, this cycle will iteratively convert SAM to SAH (Fig. 2*E*, group d). Although MeHg is too toxic to perform multiple turnover HgcAB-catalyzed reactions between SAM and Hg^{++}, these combined experiments demonstrate conclusively that SAM is the methyl donor to the cobalamin center of HgcAB to generate SAH and methylcob(III)alamin, which in the presence of Hg^{++} would then undergo demethylation to generate MeHg.

Spectroscopic Characterization of the SAM-Dependent HgcAB Reaction. With the purified protein complex, we spectroscopically and kinetically characterized only the first half-reaction

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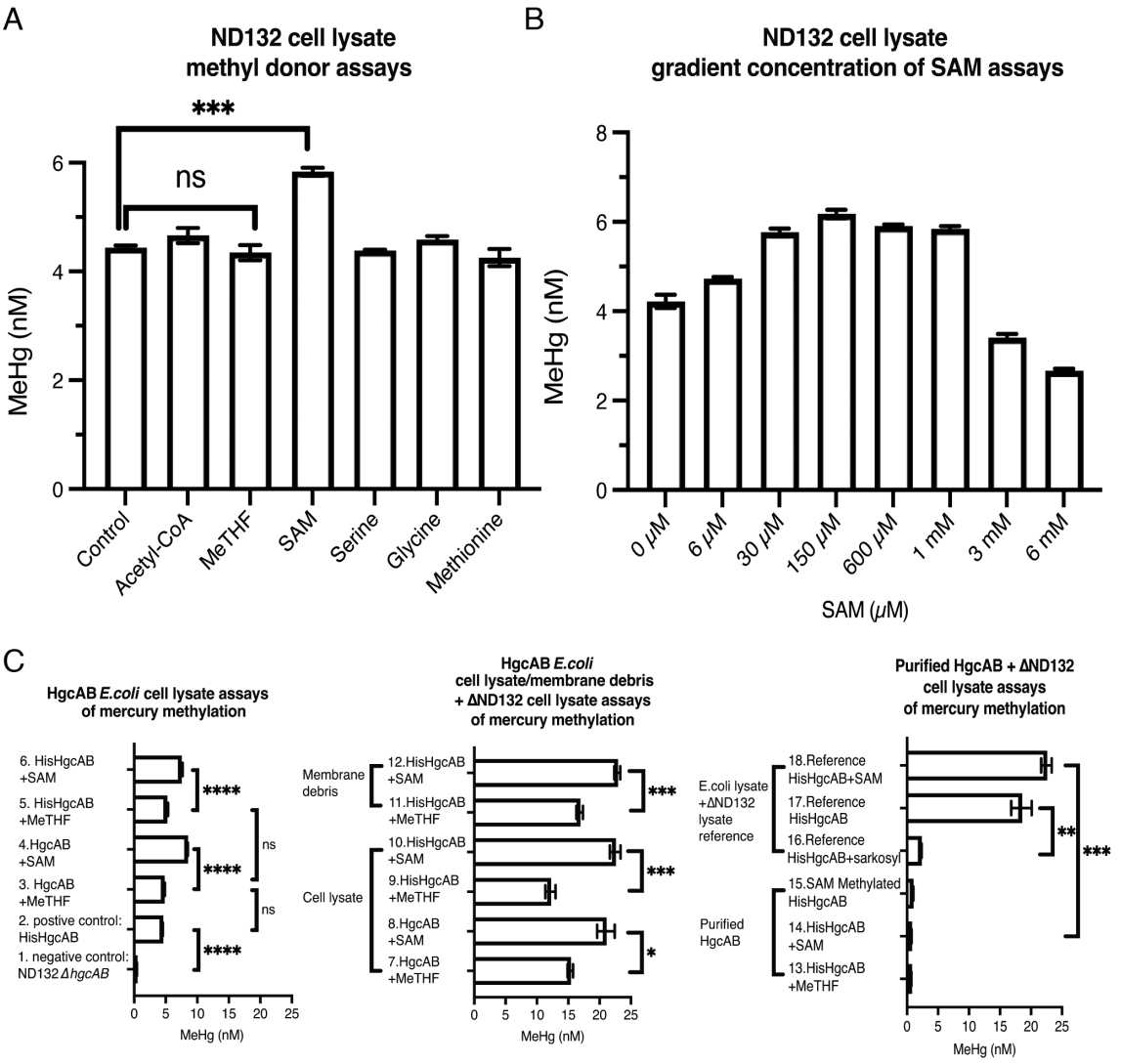


Fig. 1. Hg methylation assays using *P. mercurii* ND132 cell lysate with different methyl donors and using *E. coli* BL21 cell lysates containing HgcAB. (*A*) Mercury methylation assays for *P. mercurii* ND132 cell lysate with various methyl donors. (*B*) Mercury methylation results for *P. mercurii* ND132 cell lysate with increasing concentrations of SAM. (*C*) The *Left* panel (rows 1 to 6) presents results of Hg methylation assays using *E. coli* BL21 cell lysates containing HgcAB or N-terminal His_6-tagged HgcAB incubated with SAM or MeTHF. Row 1: negative control for Hg methylation assays using only ND132(Δ*hgcAB*) cell lysate. Row 2: positive control for Hg methylation assays using *E. coli* BL21 cell lysates containing HgcAB. Rows 3, 5: *E. coli* BL21 cell lysates containing HgcAB or N-terminal His_6-tagged HgcAB incubated with Me-THF. Rows 4, 6: Hg methylation assays performed as in rows 3, 5 except using SAM as methyl donor. The *Middle* panel presents results of Hg methylation assays using *E. coli* BL21 cell lysates containing HgcAB or N-terminal His_6-tagged HgcAB + *P. mercurii* ND132 (Δ*hgcAB*) cell lysates incubated with SAM or MeTHF. Rows 7 to 10: Hg methylation assays performed as in rows 3 to 6 but included *P. mercurii* ND132 (Δ*hgcAB*) cell lysates. Row 11 to 12: Assays as in rows 9 and 10, respectively, except using *E. coli* BL21(DE3) membrane debris separate from *E. coli* BL21-expressed N-terminal His_6-tagged HgcAB. The *Right* panel presents results of Hg methylation assays using purified N-terminal His_6-tagged HgcAB and reference controls, including *E. coli* BL21 cell lysates + *P. mercurii* ND132 (Δ*hgcAB*) cell lysates containing N-terminal His_6-tagged HgcAB incubated ±SAM or ±Sarkosyl. Rows 13 to 14: Hg methylation assays with purified HgcAB and ND132(Δ*hgcAB*) cell lysate with SAM/MeTHF. Row 15: SAM methylated purified HgcAB were incubated with ND132(Δ*hgcAB*) cell lysate, and mercury methylation assays were performed. Row 16: *E. coli* BL21 cell lysates containing N-terminal His_6-tag HgcAB were incubated with Sarkosyl and *P. mercurii* ND132 (Δ*hgcAB*) cell lysates, and mercury methylation assays were performed. Row 17: Same as row 16, except with Sarkosyl. We conducted the assays described in rows 16 and 17 because we use Sarkosyl in our purification procedure, and these assays revealed that Sarkosyl disrupts the second half-reaction. Row 18: Same as Row 10. All *P* values were obtained through a *t* test using GraphPad Prism 9. Error bars represent SD. N = 3 for each bar. ns: $P > 0.05$; *$P \le 0.05$; **$P \le 0.01$; ***$P \le 0.001$; ****$P \le 0.0001$. Details of assay conditions are described in *SI Appendix*, Table S2. Experimental details are provided in the *Materials and Methods* and *SI Appendix*.

(methylation of the Co center of HgcA), because potentially toxic levels of MeHg could have formed had we performed the full reaction. Thus, we monitored HgcAB methylation by single-turnover UV-visible (UV-Vis) (Fig. 2 *A*–*D*), EPR (Fig. 3 *A* and *B*), X-ray absorption (XAS) (Fig. 4) spectroscopies and transient stopped-flow kinetics (*SI Appendix*, Fig. S8). Upon treatment of the purified HgcAB complex (red, Fig. 2*A*; black, Fig. 2 *B* and *C*) with a ninefold excess of Ti(III)-citrate (Fig. 2 *B* and *C*), the characteristic 390 nm peak of cob(I)alamin slowly appeared (36). Addition of ninefold excess Me-THF in the dark did not cause any obvious UV-vis spectral changes even after 20 min (Fig. 2*B*). However, upon addition of a ninefold excess of SAM in the dark, the 390 nm Co(I) peak disappeared within ~20 s (Fig. 2*C*), as a broad 479 nm peak (Fig. 2*D*) appeared, indicating the consumption of cob(I)alamin and formation of methylcob(III)alamin. This reaction occurred >50-fold faster than SAM methylation of free cob(I)alamin, which takes ~20 min under these conditions (*SI Appendix*, Fig. S5).

In addition to the cobalamin spectra, we observed a 420 nm peak whose intensity varied slightly among different purification batches. We conducted pyridine hemochromagen and TMBZ heme gel staining assays (37), confirming the presence of a low level (10 to 15%) of heme copurifying with the HgcAB complex. As shown in the difference spectra in the Fig. 2, *Insets*, there is no change in the Soret

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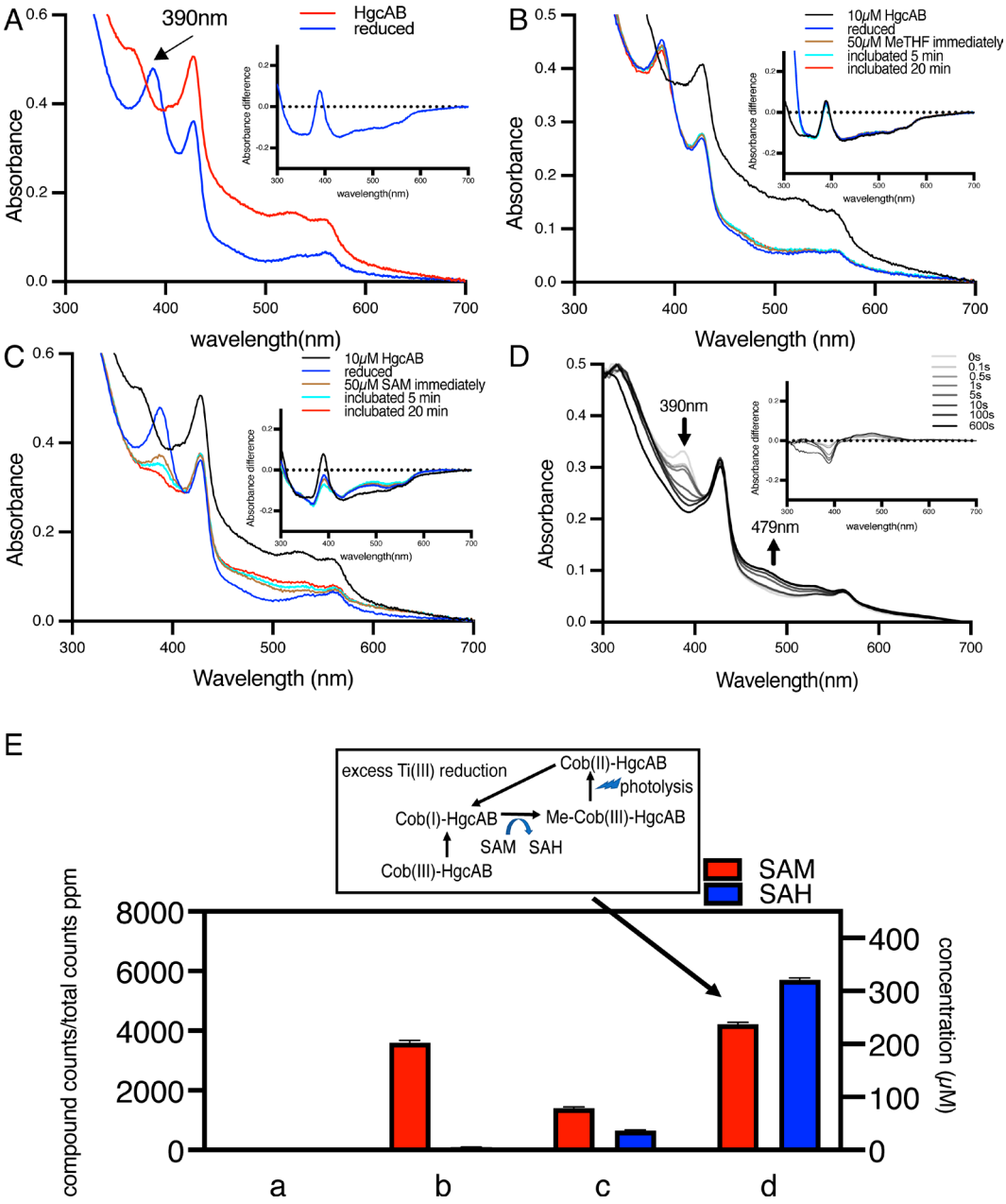


Fig. 2. UV-visible spectroscopic and LC–MS characterization of HgcAB methylation. (*A*–*D*) UV-vis spectra of (*A*) 10 μM Cob(III)-HgcAB before and after addition of 100 μM Ti(III) citrate reductant to obtain Cob(I)-HgcAB. (*B*) 10 μM Cob(III)-HgcAB and after reduction to Cob(I)-HgcAB as in *A*; then, after addition of 50 μM Me-THF at 0, 5, and 20 min. (*C*) 10 μM Cob(I)-HgcAB formed as in *A* and at 0, 5, and 20 min after addition of 50 μM of SAM. (*D*) 5 μM Cob(I)-HgcAB after reaction with 50 μM of SAM for 0, 0.1, 0.5, 1, 5, 10, 100, and 600 s. *Insets* in each figure show the UV-vis spectral differences between reduced HgcAB or methylated HgcAB and initial HgcAB spectra [Cob(III)-HgcAB or Cob(I)-HgcAB]. (*E*) Triple quadruple time of flight liquid chromatography/mass spectrometric (Q-TOF LC/MS) identification of SAH as product of the SAM methylation reaction, as in "d". (a) 50 μM Cob(I)-HgcAB, (b) quenched "a" + 200 μM SAM, (c) "a"+ 150 μM SAM, and (d) "a" + 400 μM SAM, under photolysis conditions to enable multiple turnovers as described in the scheme above the data and the main text. The *Y* axis represents SAM or SAH (ppm peak height from the LC–MS signal, and μM SAM remaining or SAH generated). Error bars represent the SD (N = 3). Other details are provided in the *Materials and Methods* and *SI Appendix*.

band intensity or position during the HgcAB SAM methylation assays, suggesting that the heme does not play a role in Hg methylation by HgcAB and that it may merely be a contaminant of protein expression in *E. coli*. The heme appears to be tightly bound and could not be removed from the HgcAB complex by incubation with apomyoglobin; furthermore, heme addition to the purified protein did not increase its levels. Additional studies are needed to firmly delineate any potential role of this heme ligand in Hg methylation.

To characterize the electronic states of B_{12} in HgcA during SAM-dependent methylation, we performed EPR spectroscopy at 60 K, which selectively visualizes cob(II)alamin, since the $[4Fe\text{-}4S]^{1+}$ clusters relax too quickly to be observed above 20 K (38). Anaerobically purified HgcAB is EPR silent; however, after reduction by 2 mM DTT, an eight-line cob(II)alamin (S = 7/2, $A_{||}$ = 106 G) EPR signal appeared with a characteristic three-line superhyperfine pattern (A_N= 17G) (Fig. 3 *A* and *B*), indicating axial N-coordination to the cob(II)alamin. These spectral characteristics resemble those of methionine synthase (39, 40) and other base-on (or His-on) cob(II)alamins, instead of base-off/His-off systems like the *Moorella thermoacetica* corrinoid iron–sulfur protein (CoFeSP) (41) or Cys-ligated cob(II)alamin, which we expected based on prior modeling results (14). The EPR spectra of Ti(III)-reduced HgcAB (both before and after addition of SAM) include a strong EPR signal at 3,400 G reflecting the excess Ti

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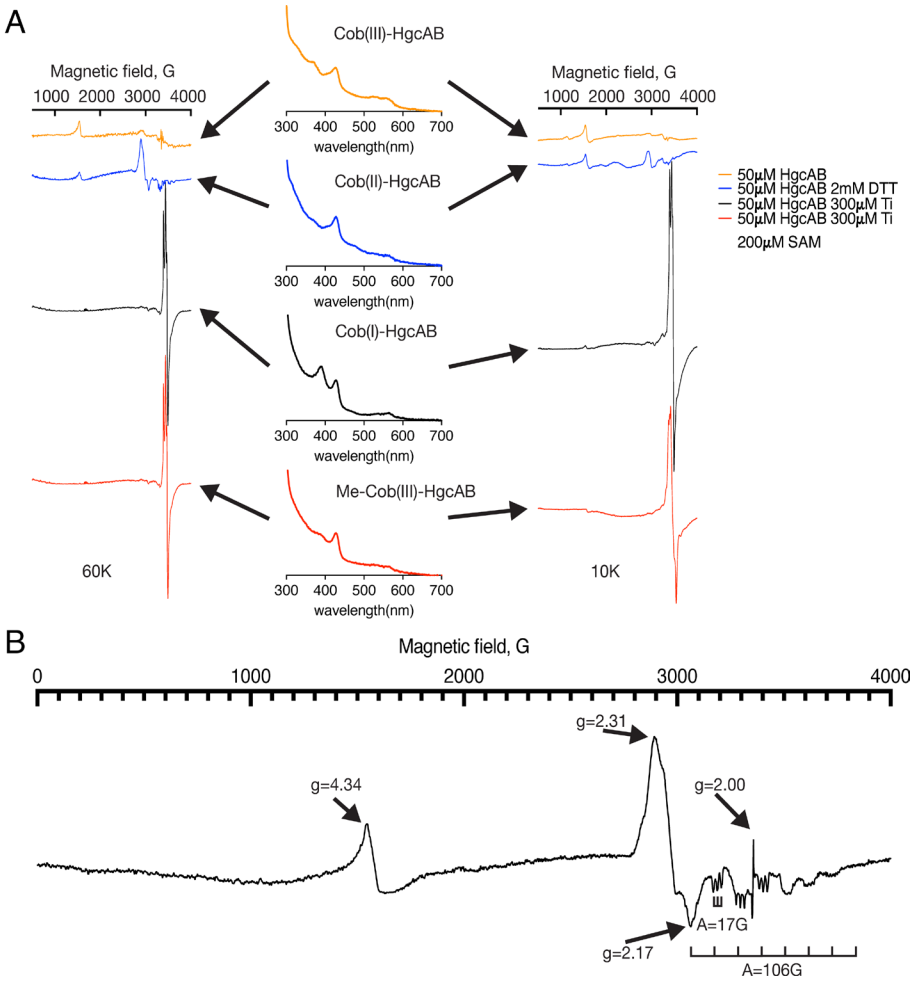


Fig. 3. EPR spectroscopic characterization of the HgcAB methylation reaction. (*A*) EPR spectra at 60 K (*Left*) and 10 K (*Right*) of Cob(III)-HgcAB, Cob(II)-HgcAB, Cob(I)-HgcAB, and SAM-generated Me-Cob(III)-HgcAB. The middle figures show the UV-vis spectra associated with the corresponding EPR spectra on the *Left* and *Right*. (*B*) Detailed view of Cob(II)-HgcAB EPR spectrum, which indicates its base-on, Cys-off coordination mode. For further details, see the *Materials and Methods*.

citrate; however, neither of these spectra contain features of cob(II)alamin, suggesting the presence of cob(I)alamin and methyl-cob(III)alamin, respectively, in these samples. Based on the EPR and UV-visible experiments, it appears that as-isolated HgcA contains cob(III)alamin (with a sulfur or nitrogen ligand), which undergoes mild reduction to an N-coordinated cob(II)alamin (Fig. 5). This state could appear as a catalytic intermediate if the methyl group transfers to Hg^{++} as a radical; on the other hand,

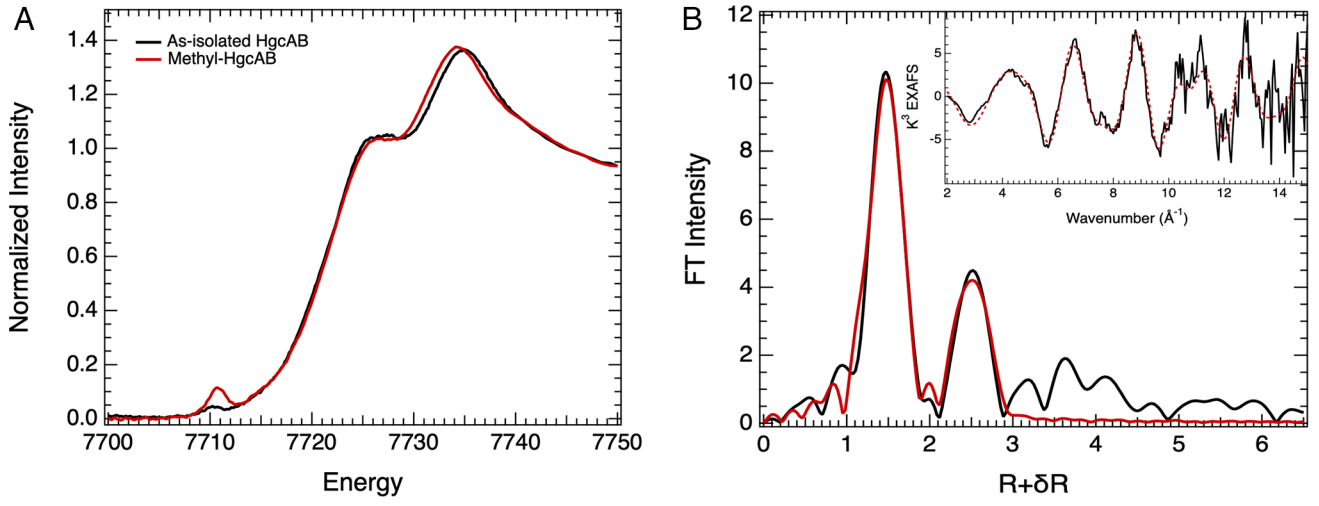


Fig. 4. Co K-edge XAS of HgcAB. (*A*) Co K-edge XAS of methylated (red) and as-isolated (black) HgcAB. (*Inset*) Expanded Co K-pre-edge region. (*B*) Non-phase-shift-corrected Fourier transform and (*Inset*) corresponding EXAFS for methylated-HgcAB. Data (black), fit (red). For further details, see the *Materials and Methods* and *SI Appendix*.

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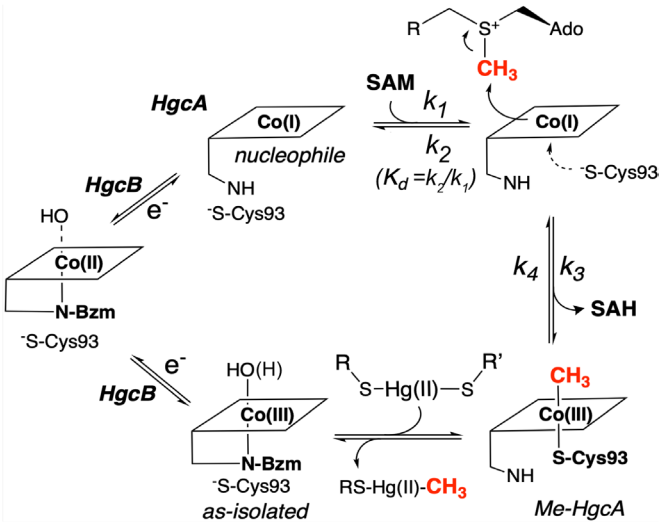


Fig. 5. Proposed mechanism for enzymatic mercury methylation catalyzed by HgcAB. For the as-isolated Co(III) state, the upper axial ligand is likely to be water or OH. Two-electron reduction by the two ferredoxin-like $[Fe_4S_4]^{1+}$ clusters of HgcB yields the nucleophilic Co(I) state, which attacks the electrophilic methyl group of SAM to generate methyl-Co(III)-thiolato cobalamin, which likely transfers its methyl group as a methyl radical (or methyl carbanion) to Hg(II) to generate MeHg.

Co(II) may only appear as an inactive species formed by oxidation of Co(I), as also observed in other cobalamin-dependent methyltransferases, which require reductive activation to reenter the catalytic cycle (42). The as-isolated or Co(II) states undergo low-potential reduction to cob(I)alamin, which is activated for SAM-dependent methylation to form methylcob(III)alamin.

To further examine the redox and ligation states of the metallocofactors in HgcAB, we performed XAS studies. Co K-edge XAS were measured on the as-isolated and methylated (Me-HgcAB) forms of HgcAB (Fig. 4*A*). The rising-edge energies (measured at 0.5 normalized intensity) are similar for both species and agree with previously reported cob(III)alamin species (43). The *Inset* of Fig. 4*A* shows the expanded pre-edge region, which represents the Co 1s → 3d transition and gains intensity through symmetry-allowed Co 3d-4p mixing. The pre-edge intensity of as-isolated HgcAB is characteristic of a typical octahedral Co(III) complex and supports the presence of weak axial ligands. Voigt fits reveal a 0.5 eV shift in energy and a ~3.7-fold increase in intensity in this feature upon methylation (Table 1). This increase is characteristic of a short axial Co-C bond, which leads to distortion in Co centrosymmetry and an increase in 3d-4p mixing. In related systems, the absolute intensity of this feature is proportional to the Co-C bond distance. *SI Appendix*, Fig. S6*A* and its *Inset* compare Me-HgcAB with methyl-cob(III)alamin (Me-Cbl) and methyl-co(III)binamide (Me-Cbi). Me-Cbl is in the base-on conformation, with a trans-axial Co-N_{DMB} ligand, while Me-Cbi lacks a tightly bound trans-axial ligand, which significantly increases the pre-edge intensity. The similarity in pre-edge intensity with Me-Cbl indicates the presence of a trans-axial ligand in Me-HgcAB.

Best fits to the non-phase-shift-corrected Fourier transforms and Extended X-ray Absorption Fine Structure (EXAFS) data of Me-HgcAB using Feff (Fig. 4*B* and Table 1) reveal the presence of Co-C coordination at 2.02 Å and a long Co-S contribution at 2.5 Å. Me-Cbl fits, performed as a reference for Me-HgcAB, (Table 1 and *SI Appendix*, Fig. S6*B*) are consistent with a shorter Co-C bond at 1.98 Å and a Co-N contribution at 2.16 Å. A Co-S interaction similar to Me-HgcAB was observed in a different cobalamin cofactor (44), although the species was not methylated. These data reveal the formation of a trans-axial Co-S(Cys) bond in a methylated cobalamin cofactor. EXAFS data for the as-isolated species were obtained to lower k and were well modeled with a six-coordinate species with two axial Co N/O ligands at 2.39 Å (*SI Appendix*, Fig. S6*C*). DFT and TD-DFT calculations were performed on a six-coordinated model with N(DMB)/OH_2 axial coordination (Table 1), which structurally reproduce the EXAFS data. Co K-pre-edge TD-DFT calculations were also consistent with the experimental data (*SI Appendix*, Fig. S6*D*).

DFT geometry optimization of a methylated Cob(III) species with an axially bound S(Cys) yields a Co-C distance of 2.01 Å and a Co-S at 2.47 Å (Table 1 and *SI Appendix*, Fig. S6*B*), consistent with the EXAFS results. Furthermore, DFT reveals that the replacement of N_{DMB} by S(Cys) results in an elongation of the Co-methyl bond by ~0.03 Å. This elongation contributes to the decrease in the XANES pre-edge intensity of the Me-HgcAB relative to Me-Cbl (*SI Appendix*, Fig. S6 *A*, *Inset*), owing to a decrease in Co 4p character in the states that contribute to the pre-edge feature in Me-HgcAB, which is reproduced well by TD-DFT calculations (*SI Appendix*, Fig. S6*E*). These results indicate the presence of a long Co-S bond, supporting a Co-S(cys) coordination in Me-HgcAB. The difference in Co-C bond length relative to Me-Cbl suggests that the unprecedented Co-S interaction in a methylated cobalamin cofactor might be designed to tune the Co-C bond and poise it for Hg methylation.

The Co XAS data of the Cys-on methylated state are consistent with a prior study by Li et al. who observed a Co-S(Cys) bond at a distance of 2.56 Å in the base-off, unmethylated B_{12} trafficking protein CblD (44). This is an experimental validation of a Co-S assignment originally predicted in 2013 (3) and later supported by quantum chemical calculations performed by us (24) and by others (23) as well as using site-directed mutagenesis and in vivo Hg methylation experiments (3, 18). These latter experiments described the critical roles of Cys93 in HgcA and Cys73, Cys94,

Table 1. DFT and TD-DFT geometry optimization calculations on the Me-Co(III) states of HgcAB, methylcobalamin, and aquocobalamin related to the XAS data described in Fig. 4 and *SI Appendix*, Fig. S6*

	Methyl-Cob(III)-Cys	Methylcobalamin	Aquocobalamin
Co-N(Corrin)	1.924*	1.914*	1.883*
	1.875	1.873	1.926
	1.876	1.871	1.920
	1.936	1.922	1.883
Average Co-N (corrin)	1.903	1.895	1.903
Coaxial ligand (methyl or water)	2.007	1.974	2.050
Co-trans-axial ligand (S, DMB, water)	2.469	2.153	1.916

*All distances in Å.

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and Cys95 in HgcB for Hg methylation. It is important in future investigations to determine whether the Co-S bond is most important for methylation of Co in the first half-reaction or for Hg in the second half-reaction.

Kinetic Studies of the SAM-Dependent Methylation of B_{12} by HgcAB. We propose that the first half-reaction, generation of methyl-Co(III), involves transfer of a methyl cation (Fig. 5). We also suggest that the second half-reaction is likely linked to a unique radical- or anion-based mechanism that may involve the conserved C-terminal cysteines (C94 and 95) in HgcB, for Hg transfer from solution and/or for reaction with the Co-S center (involving Cys93) to yield MeHg. Cobalamin methylation by HgcAB occurs 50 times faster than the analogous SAM-dependent methylation of TrsM, GenK, Fom3, and PhpK, and of free Cob(I)alamin (45–49); thus, we characterized the kinetics via stopped-flow UV-vis spectroscopy. We mixed 10 μM Ti(III) citrate-reduced HgcAB with increasing concentrations of SAM and observed that decay of the 390 nm Co(I) absorption follows a double exponential progress curve, matching the increase at 479 nm (*SI Appendix*, Fig. S7). The stopped-flow data were acquired on a logarithmic time scale and analyzed by fitting the data to a double exponential equation (Eq. **2**) for both 390 nm and 479 nm (*SI Appendix*, Fig. S7), yielding rate constants k_{fast} and k_{slow} (Eq. **2**) (*SI Appendix*, Table S4), which correspond to the fast and slow changes in the 390 nm and 479 nm signals. When k_{fast} or k_{slow} were plotted against SAM concentration, a saturation curve was observed for k_{fast} (*SI Appendix*, Fig. S8*D*), while k_{slow} remained SAM independent. Additionally, a comparison of the UV-vis difference spectra associated with k_{slow} or k_{fast} revealed significant differences (*SI Appendix*, Fig. S9). The amplitude of the 390 nm peak for k_{fast} accounted for 75% of the total and was associated with the increase at 479 nm. Conversely, the broad shoulder increase spanning from 450 nm to 520 nm was associated with k_{slow}. These observations suggest that k_{slow} corresponds to a side reaction(s), likely unrelated to Cob(I) methylation (*SI Appendix*, *Supplementary Materials* and Figs. S7 and S9). These side reaction(s) may involve nonenzymatic oxidation of Co(I) or slow homolytic cleavage of methyl-Co(III) to generate Co(II). Thus, we focused our attention on k_{fast}.

We propose a two-step reversible mechanism described by Eq. **1** ($K_d = k_2/k_1$). The traditional nonlinear fitting approach for acquiring kinetic parameters involves many approximations, such as square root approximations and other hypothetical assumptions, to simplify complex equations that are difficult to fit. These approximations can introduce errors, resulting in kinetic parameters that deviate significantly from the true values. Using simulation and fitting through kinetics software is an effective way to mitigate this issue. Therefore, we built a double-step reversible mechanism model in KinTek Explorer software and fitted our kinetic data with the model to acquire accurate kinetic parameters. To prevent interference from nonenzymatic oxidation of Co(I) or the slow homolytic cleavage of methyl-Co(III) generating Co(II), we selected 390 nm and 479 nm kinetics data within the 0.05 s to 5 s range. The k_1 and k_2 values for the forward and reverse reactions of the first step are 0.55 ± 0.04 $\mu M^{-1}\cdot S^{-1}$ and 16 ± 1 s^{-1}, and the k_3 and k_4 values for the forward and reverse reactions of the second step are 0.160 ± 0.003 s^{-1} and 0.002 ± 0.001 $\mu M^{-1}\cdot s^{-1}$, according to the 390 nm kinetics data (*SI Appendix*, Fig. S8*A*). When fitting the model using the 479 nm data, k_1, k_2, k_3, and k_4 are 0.40 ± 0.09 $\mu M^{-1}\cdot s^{-1}$, 16 ± 4 s^{-1}, 0.13 ± 0.02 s^{-1}, and 0.002 ± 0.01 $\mu M^{-1}\cdot s^{-1}$, respectively (*SI Appendix*, Fig. S8*B*). The global fitting, which includes both 390 nm and 479 nm kinetics data, yielded values of k_1, k_2, k_3, and k_4 of 0.48 ± 0.09 $\mu M^{-1}\ddot{\imath}\cdot s^{-1}$, 16 ± 3 s^{-1}, 0.15 ± 0.02 s^{-1}, and 0.002 ± 0.009 $\mu M^{-1}\cdot s^{-1}$ (*SI Appendix*, Fig. S8*C*). K_D(SAM) thus equals 33 μM. The 50 to 75-fold larger rate of SAH formation (k_3) relative to that of reformation of SAM (k_4), indicates that the SAM-dependent methylation of HgcA is irreversible.

$$\mathrm{Cob(I)-HgcAB + SAM} \underset{k_2}{\overset{k_1}{\rightleftharpoons}} \mathrm{Cob(I)-HgcAB\ SAM} \underset{k_4}{\overset{k_3}{\rightleftharpoons}} \mathrm{Me-Cob(III)-HgcAB + SAH}, \quad [1]$$

$$A = ae^{k_{fast}t} + be^{k_{slow}t} + c. \quad [2]$$

Discussion

It has been three decades since the discovery of the enzymatic basis for microbial mercury methylation. Previous mercury methylation studies using isotope labeling mass spectrometry identified an association between the Wood–Ljungdahl pathway and mercury methylation, suggesting Me-THF as the methyl donor (11–13). Subsequent sequence analyses reinforced this hypothesis because of the homology between the CoFeSP involved in acetyl CoA synthesis and the cobalamin binding domain of HgcA (3). Furthermore, metabolic studies and sequence comparisons implicated Me-THF as the methyl donor for mercury methylation; however, no direct evidence for these assertions has been reported due in part to the low levels of HgcAB expression in microorganisms and the difficulties of purifying this O_2-sensitive, membrane-associated protein complex.

Here, we propose a mechanism for HgcAB-dependent mercury methylation that uses SAM as a methyl donor (Fig. 5). Based on our spectroscopic studies, the resting state of the mercury methylation complex is Cob(III)-HgcAB. It is unclear whether the upper axial ligand is water or OH^-, and the pK_a for aquo-Cob(III)-alamin is in the pH 6-8 range. Correlation of the pre-edge XAS spectra with TD-DFT agrees with this assignment, indicating a tetragonally elongated Cob(III) species with weakly bound water or OH. Since Cob(III) is low-spin d_6, such a distortion would yield vacant dx^2-y^2 and d_z^2 states, which can be reduced by a weak reductant such as dithiothreitol (DTT) and form an EPR-active N-coordinated Cob(II)-HgcAB. The octet Co(II) and triplet N-splittings are similar in magnitude to those seen in the base-on states of methionine synthase (42), reflecting either Co(II) coordination of the N-benzimidazole attached to the cobalamin ring or perhaps a His residue. Based on our prior modeled structure of HgcA, there are no nearby His residues (14), favoring benzimidazole as the axial ligand; however, further studies will be required to identify the axial Co(II) ligand.

During the catalytic cycle, HgcAB is reduced to the Cob(I)-HgcAB state through electron transfer mediated by the [4Fe-4S] clusters in HgcB. These two clusters are unlikely to function in 5-deoxyadenosyl radical generation like class B radical SAM enzymes, which only coordinate with three cysteine ligands (50). In these site-differentiated [4Fe-4S] clusters, the differentiated iron does not coordinate with cysteine and can interact with SAM to generate a 5-deoxyadenosyl radical. This radical then initiates

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catalysis by substrate H atom abstraction at the eventual methylated position, with the methyl group originating from another molecule of SAM via intermediate methylation of the cobalamin cofactor. Instead, based on sequence analysis and structural modeling, both [4Fe-4S] clusters in HgcB have four associated cysteine ligands (3, 14), homologous to typical 8-Fe ferredoxins (51, 52). Thus, we propose that both [4Fe-4S] clusters in HgcB serve as electron transfer mediators. These two [4Fe-4S] clusters in HgcB exhibit complex EPR spectra when they are both reduced (*SI Appendix*, Fig. S4), indicating they are spaced close enough (<14 Å) to facilitate rapid electron transfer to regenerate Co(I) during reactive activation and catalysis. The origin of these two reducing equivalents is unknown. We suggest that pyruvate-ferredoxin oxidoreductase, hydrogenase, or other low-potential electron donors could serve this role.

After two-electron reduction to Cob(I)-HgcAB, a molecule of SAM can bind to the HgcAB complex and transfer its methyl group to Cob(I)-HgcA as a methyl cation with the generation of one molecule of SAH. The Cys thiolate ligand then coordinates as a lower axial ligand to stabilize Me-Cob(III)-HgcAB. Then, Hg^{++} is delivered as $Hg(RS)_2$ to the active center of HgcAB and the methyl group from Me-Cob(III)-HgcAB is transferred to $Hg(RS)_2$ as a methyl radical or carbanion, forming MeHgSR and another thiolate (RS^-), reverting HgcAB back to its Cob(III)-HgcAB resting state.

Questions about the mechanistic details of the second half-reaction remain. Conducting the full Hg methylation assay with purified, active enzyme will generate large quantities of neurotoxic methylmercury. It is also unknown whether the enzyme can accept monomethylmercury as a substrate, and potentially generate the even more deadly dimethylmercury (53). Figuring out how to safely conduct this experiment, or validating some alternative less dangerous assay, will be a significant future undertaking.

Our findings conclusively demonstrate that SAM, not Me-THF, serves as the source of the methyl group in MeHg, further expanding the biochemical role of SAM. The understanding of SAM-dependent enzymes already has undergone a major expansion over the past two decades, with the discovery of radical SAM enzymes (54). A variety of SAM- and cobalamin-dependent reactions are catalyzed by class B radical SAM enzymes; examples include TokK, Fom3 (45, 55–57), and Mmp10 (58–60). TsrM, which catalyzes methylation of posttranslationally modified peptides (RiPPs) is an atypical member of this class in that it does not appear to use a dAdo radical, as we propose for HgcAB (36, 49). Additionally, we are not aware of any other class B radical SAM enzyme and corrinoid iron–sulfur methyltransferase that contains an essential transmembrane domain. The involvement of SAM in HgcAB-catalyzed Hg methylation also distinguishes HgcAB from the CoFeSP involved in B_{12} and Ni ion methylation in the Wood–Ljungdahl pathway.

In summary, demonstrating SAM as the methyl donor in MeHg formation by the HgcAB enzymatic system overturns a paradigm linking this process to Me-THF. MeHg perversely biomagnifies through the food web to become a potent human neurotoxin. Significant challenges of low expression levels and poor cofactor incorporation were overcome to allow enzymatic and biophysical characterization of this integral membrane-associated B_{12}- and Fe_4S_4-dependent system. Unusual features of the HgcA active site include formation of a Me-B_{12} intermediate with a cysteine thiolate serving as a lower axial cobalt ligand. These findings may promote development of strategies to mitigate against the microbial production of toxic Me- and diMe-Hg in the environment.

Materials and Methods

If not specified, all chemicals were commercially purchased from Sigma-Aldrich, St. Louis, MO. In all assays of the first half-reaction of HgcAB (methylation of the cobalamin in HgcA), we were cautious to ensure that no forms of Hg are present. Full Experimental Details for protocols and reagents related to the Materials and Methods are provided in *SI Appendix*.

Cloning, Expression, and Purification of HgcAB. To overcome the low native protein levels, we cloned the *hgcA* and *hgcB* genes from *P. mercurii* (formerly *D. desulfuricans ND132*) in tandem into *E. coli* pETDuet-1 expression vector (Novagen, Madison, WI). Sequences were verified by Sanger sequencing (University of Michigan DNA Sequencing Core). When expressed independently of HgcB, HgcA was unstable and insoluble, lacked cobalamin, and had very low expression levels. Similarly, HgcB expressed without HgcA was insoluble. However, expression of HgcA and HgcB in the tandem cloning sites of the pET duet vector produced a soluble HgcAB complex.

To ensure that the two [4Fe-4S] clusters were incorporated into HgcB and the cobalamin center into HgcA, we cotransformed the pRKisc vector for [4Fe-4S] cluster synthesis (29–31) and the pBtu vector for cobalamin uptake (32) with the HgcAB-containing pETDuet-hgcA-hgcB or pETDuet-His-hgcA-hgcB vector into both *E. coli* BL21(DE3) and C41(DE3) strains (28). In cell lysate assays for mercury methylation, constructs cloned into BL21(DE3) exhibited mercury methylation activity, while those cloned into C41(DE3) did not (not shown). Thus, we used the BL21(DE3) strain as our expression platform. After performing parallel experiments in BL21(DE3) under different conditions, we observed that the N-terminal His-tag does not significantly influence the HgcAB mercury methylation activity (compare rows 3 & 5 in Fig. 1*C*).

Anaerobic expression of HgcAB in M9-ETA media (components listed in *SI Appendix*, Table S1) provided the highest activity (*SI Appendix*, Fig. S1 and Table S3). Thus, we selected this growth condition to express the N-terminal His-tagged HgcA with a nontagged HgcB for further purification and characterization. We used a 10 L fermenter for HgcAB expression. Use of an N-terminal His_6-tag on HgcA enabled purification of the HgcAB complex using Ni-NTA agarose chromatography (33). Because HgcA is a transmembrane protein, detergents were used to maintain solubility of the HgcAB complex. Our optimized purification protocol included 0.3% Sarkosyl and 0.6% Triton X-100, which enabled elution of homogeneous HgcAB from the Ni-NTA agarose.

All purification steps were performed in an anaerobic chamber that maintained O_2 levels below 1 ppm. The cell pellets obtained from fermentation were resuspended in anaerobic Buffer A (50 mM K_2HPO_4, 100 mM NaCl, 10% glycerol, and 1 mM TCEP, pH 7.4) containing protease inhibitor and 10 mM imidazole (10 mL buffer/g cell pellets). Next, the resuspended cell pellets were sonicated and the cell lysates were anaerobically centrifuged. The supernatants were discarded, and the precipitants containing the membrane debris were weighed and resuspended (0.13 g/mL) in Buffer A with 1% Sarkosyl, 2% Triton X-100, 10 mM imidazole for 1 h at room temperature and sonicated, resuspended, and resonicated. The sonicated mixture was anaerobically centrifugated and the supernatant was applied to a Ni-NTA agarose column pre-equilibrated with Buffer A containing 0.3% Sarkosyl, 0.6% Triton X-100, washed, and eluted with Buffer A containing 0.15% Sarkosyl, 0.15% Triton X-100, 250 mM imidazole. The resulting HgcAB complex was typically >95% pure, containing the expected 37 kDa HgcA and the 11 kDa HgcB Coomassie-stained protein bands by SDS-PAGE (*SI Appendix*, Fig. S2).

The HgcAB complex was buffer exchanged using an Amicon centrifugal filter unit (Amicon® Ultra - 0.5 mL 30 kDa, Amicon, Miami, FL) and dialyzed overnight against Buffer B (20 mM HEPES, 150 mM NaCl, 1 mM TCEP, and 10% glycerol, pH 7.4). To prepare HgcAB samples for XAS characterization, an additional ethylenediaminetetraacetic acid (EDTA) dialysis step was included to remove Ni contamination that interferences XAS signal. After buffer exchange using an Amicon centrifugal filter unit, HgcAB samples were dialyzed overnight against Buffer C (20 mM HEPES, 150 mM NaCl, 1 mM TCEP, 1 mM EDTA, 0.1% Sarkosyl, and 10% glycerol, pH 7.4). Then, the HgcAB samples were dialyzed overnight against Buffer B (20 mM HEPES, 150 mM NaCl, 1 mM TCEP, and 10% glycerol, pH 7.4). The purified HgcAB complex stoichiometrically bound cobalamin (34, 35) (*SI Appendix*, Fig. S3 *A* and *B*) and two [4Fe-4S] clusters. MS/MS identification of SDS-PAGE gel slices validating the identity of the 37 and 11 kDa bands (*SI Appendix*, Fig. S2), and purified protein provided ~90% coverage for both HgcA and HgcB (*SI Appendix*, Fig. S3*C*).

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***P. mercurii* Culture Conditions and Preparation of Cell Lysates.** WT *P. mercurii* or Δ*hgcAB* cultures (ND132 cells lacking the *hgcA* and *hgcB* genes) were prepared as described previously (25) and were grown in modified minimal organic basal medium with yeast extract (MOY) (18) containing 40 mM fumarate, 40 mM pyruvate, and 1.2 mM thioglycolate and 1 mM cysteine·HCl as reducing agents for approximately 3 d (OD_{600} = 0.3) at 32 °C under anaerobic conditions. Cell pellets were harvested at 4 °C by centrifugation for 20 min at 7,500 × g in centrifuge bottles with sealing closures. Then, the cell pellets were washed two times with deoxygenated phosphate-buffered saline (PBS), pH 7.0, and resuspended in cold lysis buffer containing Pierce™ EDTA-free protease inhibitor and 2 mM DTT in PBS, pH 7.0, and disrupted on ice by sonication under strictly anaerobic conditions inside a glove box under an N_2 atmosphere ($O_2 \leq 1$ ppm) with minimal exposure to ambient light. Unlysed cells were removed by centrifugation at 30,000 × g for 1 h at 4 °C. Total protein concentration of cell lysates was determined by the Bradford method with bovine serum albumin (BSA) (Millipore Sigma, St. Louis, MO) as standard and the lysates were stored in aliquots at −80 °C in amber glass vials with PTFE-lined silicone caps.

HgcAB Analytical Assays. Protein was determined using the BCA assay based on a standard curve generated by BSA. The cobalamin content of HgcAB was determined as the dicyano-Co(III)balamin complex by treatment of the protein solution with KCN at 95 °C (61) and measuring the dicyano-cob(III)alamin concentration by UV-visible spectroscopy using extinction coefficients of 26,200, 7,400, and 8,600 $M^{-1}cm^{-1}$ at 367, 540, and 580 nm (34, 35). Representative spectra of as-purified HgcAB and derivatized dicyano-cob(III)alamin are shown in *SI Appendix*, Fig. S3 *A* and *B*, respectively. The HgcAB [4Fe-4S] cluster occupancy was determined by EPR after reduction by sodium dithionite (1 mM, final) for 2 h at room temperature. The EPR spectrum (*SI Appendix*, Fig. S4) indicated a coupled $[4Fe\text{-}4S]^{+}$ system, consistent with a classic eight iron ferredoxin (51, 52). The spin concentration was measured by double integration referenced to a 1 mM copper perchloride standard.

Mercury Methylation Cell Lysate Assays. These experiments were designed to produce <50 pmol of MeHg and performed in special facilities at ORNL to ensure safety. Cell lysates of *E. coli* strains containing the overexpressed HgcAB complex were prepared from frozen washed cell pellets after resuspension in lysis buffer and sonication under anaerobic conditions, as described above. After removal of unlysed cells by centrifugation and determination of the total protein concentration. Hg methylation assays were conducted in an anaerobic glove box under a pure N_2 atmosphere ($[O_2] < 1$ ppm) in the dark. Samples were added to 4 mL amber glass vials and the total protein concentration was adjusted with anoxic PBS buffer to 1.5 mg/mL. For experiments in which cell lysates were mixed, aliquots of the *E. coli* lysates were added to a *P. mercurii* Δ*hgcAB* cell lysate, to a protein concentration of 0.1 mg/mL unless specified otherwise.

To study the effect of metabolites on Hg methylation in ND132 cell lysates, freshly prepared stocks (1.5 mM) of 5-Me-THF, S-adenosyl methionine (SAM), acetyl-CoA, L-glycine, L-methionine, or L-serine were added to Hg methylation assays at a final concentration of 30 μM and in a final volume of 1 mL, or at increasing concentrations in establishing the K_m for SAM.

In all mercury methylation assays, samples were equilibrated for 5 min before adding an aliquot of a freshly prepared $HgCl_2$ stock solution (1.5 μM) to a final concentration of 30 nM and incubated at 32 °C for 2 h, quenched by adding 0.5% (v/v) trace-metal grade H_2SO_4, and stored at −20 °C until MeHg was analyzed by inductively coupled plasma mass spectrometry (ICP-MS). MeHg was measured using a modified EPA Method 1630 via distillation and ethylation (3, 62). Modifications to the EPA methods included the use of isotope dilution with enriched stable isotopes to determine total Hg and methylmercury concentrations, and detection of Hg by ICP-MS to separate the various Hg isotopes (63).

Single Turnover Assays of HgcAB Methylation. To track the HgcAB-catalyzed cobalamin methylation reaction, we performed the enzymatic assay in Buffer B. A titanium(III) citrate (Ti(III) citrate) solution was prepared by adding 100 μL of 15% Ti(III) chloride into a premixed buffer containing 1,125 μL of 0.5 M sodium citrate and 400 μL of 1 M Tris buffer, pH 8.0 for 10 min at room temperature. An aliquot was diluted into Buffer B to 66 mM Ti(III) citrate for cobalamin reduction.

Methylation assays were performed using 10 μM HgcAB in Buffer B. Then, 100 μM Ti(III) citrate (final) was added to reduce the as-purified Cob(III)- to Cob(I)-HgcAB and appearance of the 390 nm Co(I) peak was monitored. Next, 50 μM methyl donor was added and disappearance of the 390 nm Co(I) peak and appearance of a broad 479 nm peak were monitored.

The nonenzymatic methylation of cobalamin was performed similarly using 10 μM reduced hydroxocobalamin in Buffer B. To thoroughly reduce the 10 μM of hydroxocobalamin, 400 μM Ti(III) citrate was added for a 20 min reaction, and reduction was monitored by appearance of the 390 nm Co(I) peak in the UV-vis spectrum. Meanwhile, the excess Ti(III) citrate significantly increased the baseline absorbance (*SI Appendix*, Fig. S5*A*). Then, 50 μM SAM was added and the 390 nm Co(I) peak slowly disappeared in 20 min, concomitant with the appearance of a broad 520 nm peak (*SI Appendix*, Fig. S5*B*).

HgcAB Stopped-Flow Kinetic Assays. To characterize the kinetics for the HgcAB methylation reaction, stopped-flow UV-vis experiments were performed at 21 °C within the anaerobic glove box on an Applied Photophysics (Leatherhead, England). spectrophotometer equipped with monochromator and photodiode array (PDA) detector. In one syringe (Reagent A), 10 μM HgcAB in Buffer A was reduced by addition of 100 μM Ti(III) citrate (final). The other syringe (Reagent B) contained SAM (100 μM, 200 μM, 400 μM, 800 μM, 1,200 μM, 1,600 μM, or 2,000 μM) in the same buffer. Reagents A and B were mixed via a 1:1 ratio in single-mixing mode and the experiments were performed in triplicate. UV-vis spectra were collected from 300 nm to 700 nm for analysis and fitting. An example is shown in Fig. 2*D*, in which data were acquired at 50 μM SAM after mixing and spectra from different time points of the reaction were superimposed.

The stopped-flow data were acquired using Pro-data SX software. Each set of reactions was monitored for 600 s and 1,000 spectra were collected in a logarithmic time interval. Analysis was performed using Pro-data viewer software and data at both 390 nm and 479 nm were fit to a double exponential equation (*SI Appendix*, Fig. S7) to yield both k_{fast} and k_{slow} (*SI Appendix*, Table S4). The fitting of k_{fast} obtained from both 390 nm and 479 nm versus SAM concentration was performed by in GraphPad Prism 9 (*SI Appendix*, Fig. S8*D*). The kinetic data from 0.05 to 5 s at 390 nm and 479 nm were fit to a two-step reversible model via KinTek Explorer 11.1.0 software, to obtain the rate constants k_1, k_2, k_3, and k_4. The fitting method is described in more detail in Supporting Information and the fitting results are presented in *SI Appendix*, Fig. S8 and Table S5.

UV-Visible and EPR Characterization of HgcAB Methylation. To track the cobalamin states of HgcAB during Ti(III) citrate reduction and SAM methylation, the reaction was characterized in parallel by EPR and UV-visible spectroscopy. As-purified HgcAB (50 μM) was prepared in Buffer B. For UV-vis, samples were analyzed in a 1 mm path length cuvette. The Co(II) state of HgcAB was generated by incubating 50 μM HgcAB with 2 mM DTT for 120 min. To generate Cob(I)-HgcAB, 300 μM Ti(III) citrate was added to 50 μM HgcAB in Buffer B and incubated for 150 min at room temperature and monitored by UV-visible spectroscopy to verify the reduction. Then, to generate Me-Cob(III)-HgcAB, 300 μM Ti(III) citrate was added to 50 μM HgcAB in Buffer A for 150 min at room temperature, followed by addition of 200 μM SAM (final) in the dark.

LC–MS Analyses of SAM and SAH, HgcAB, and MeHg. The HgcAB-catalyzed conversion of SAM to SAH was also monitored by LC–MS. An anaerobic sample of 50 μM HgcAB in Buffer B was reduced to the Cob(I) state by reaction with 300 μM Ti(III) citrate. Then, 150 μM SAM was added in the dark, and the reaction was quenched after 5 min by addition of Quench Solution (methanol and formic acid) to precipitate the protein, the sample was centrifuged, and the supernatant was injected onto an Agilent LC–MS to separate and quantify SAM and SAH in group c of Fig. 2*E*. Meanwhile, two different control references were prepared similarly: one lacking SAM in group a of Fig. 2*E* and another with 200 μM SAM added after quenching in group b of Fig. 2*E*. A photolysis group d was also prepared to verify the synthesis of the Me-Cob(III)-HgcAB in which an anaerobic sample of 50 μM HgcAB in Buffer A was reduced to the Cob(I)-HgcAB state by 300 μM Ti(III) citrate. Then, 200 μM SAM was added, and the Eppendorf tube was exposed to a fluorescent lamp bulb (20 W/1,200 lm). After 30 min, an additional 200 μM Ti(III) citrate and 200 μM of SAM were included and the reaction was quenched after 30 min of exposure by adding Quench Solution. The protein was precipitated, and the sample was centrifuged in group d of Fig. 2*E*. All experiments were performed in triplicate and examined by LC–MS. The peaks assigned to SAM and SAH were integrated, the ratio of the integrations of SAM/(all peaks) and of SAH/(all peaks) were calculated for each sample and presented as ppm (parts per million) in

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Fig. 2*E*. Each lane represented the average value of each group of samples with error bars representing the SD.

XAS Spectroscopy. Purified HgcAB samples were loaded into XAS cuvettes sealed with 25 μm Kapton tape, flash frozen, and stored under LN2 until analysis at the Stanford Synchrotron Radiation Lightsource on beamline 9-3. During data collection, samples were maintained at 10 K using an Oxford Instruments CF 1208 liquid helium cryostat. Data were collected at the Co K-edges, and data were collected simultaneously on a Co foil standard for energy calibration. Spectra were collected at multiple spots per sample to minimize beam-induced damage, although no beam damage was observed across multiple scans on the same sample spot. Details of the XAS data analysis are provided in *SI Appendix* section.

Data, Materials, and Software Availability. All study data are included in the article and/or *SI Appendix*.

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Author affiliations: [a]Department of Biological Chemistry, University of Michigan Medical School, Ann Arbor, MI 48109-0606; [b]Environmental Sciences Division, Oak Ridge National Laboratory, Oak Ridge, TN 37831-6038; [c]Biosciences Division, Oak Ridge National Laboratory, Oak Ridge, TN 37831-6309; and [d]Department of Structural Molecular Biology, Stanford Synchrotron Radiation Lightsource, SLAC National Accelerator Laboratory, Menlo Park, CA 94025

Author contributions: K.Z., K.W.R., S.S.D., A.J., J.M.P., and S.W.R. designed research; K.Z., K.W.R., S.S.D., A.J., J.M.P., A.S.F., M.J.A., and R.S. performed research; K.Z. contributed new reagents/analytic tools; K.Z., K.W.R., S.S.D., A.J., J.M.P., M.J.A., R.S., and S.W.R. analyzed data; S.W.R. coordinated the various research groups; and K.Z. and S.W.R. wrote the paper.

1. T. W. Clarkson, Human toxicology of Mercury. *J. Trace Elem. Exp. Med.* **11**, 303–317 (1998).
2. C. T. Driscoll, R. P. Mason, H. M. Chan, D. J. Jacob, N. Pirrone, Mercury as a global pollutant: Sources, pathways, and effects. *Environ. Sci. Technol.* **47**, 4967–4983 (2013).
3. J. M. Parks *et al.*, The genetic basis for bacterial mercury methylation. *Science* **339**, 1332–1335 (2013).
4. C. C. Gilmour *et al.*, Mercury methylation by novel microorganisms from new environments. *Environ. Sci. Technol.* **47**, 11810–11820 (2013).
5. C. C. Gilmour, A. L. Bullock, A. McBurney, M. Podar, D. A. Elias, Robust mercury methylation across diverse methanogenic archaea. *mBio* **9**, e02403-17 (2018).
6. M. Harada, Minamata disease: Methylmercury poisoning in japan caused by environmental pollution. *Crit. Rev. Toxicol.* **25**, 1–24 (1995).
7. S. C. Choi, T. Chase, R. Bartha, Enzymatic catalysis of mercury methylation by *Desulfovibrio desulfuricans* LS. *Appl. Environ. Microbiol.* **60**, 1342–1346 (1994).
8. M. Podar *et al.*, Global prevalence and distribution of genes and microorganisms involved in mercury methylation. *Sci. Adv.* **1**, e1500675 (2015).
9. E. A. McDaniel *et al.*, Expanded phylogenetic diversity and metabolic flexibility of mercury-methylating microorganisms. *mSystems* **5**, e00299-20 (2020).
10. E. Capo *et al.*, A consensus protocol for the recovery of mercury methylation genes from metagenomes. *Mol. Ecol. Resour.* **23**, 190–204 (2023).
11. M. Berman, T. Chase, R. Bartha, Carbon flow in mercury biomethylation by *Desulfovibrio desulfuricans*. *Appl. Environ. Microbiol.* **56**, 298–300 (1990).
12. S.-C. Choi, T. Chase, R. Bartha, Metabolic pathways leading to mercury methylation in *Desulfovibrio desulfuricans* LS. *Appl. Environ. Microbiol.* **60**, 4072–4077 (1994).
13. E. B. Ekstrom, F. M. M. Morel, J. M. Benoit, Mercury methylation independent of the acetyl-coenzyme A pathway in sulfate-reducing bacteria. *Appl. Environ. Microbiol.* **69**, 5414–5422 (2003).
14. C. J. Cooper *et al.*, Structure determination of the HgcAB complex using metagenome sequence data: Insights into microbial mercury methylation. *Commun. Biol.* **3**, 320 (2020).
15. T. Svetlitchnaia, V. Svetlitchnyi, O. Meyer, H. Dobbek, Structural insights into methyltransfer reactions of a corrinoid iron–sulfur protein involved in acetyl-CoA synthesis. *Proc. Natl. Acad. Sci. U.S.A.* **103**, 14331–14336 (2006).
16. Y. Kung *et al.*, Visualizing molecular juggling within a B12-dependent methyltransferase complex. *Nature* **484**, 265–269 (2012).
17. S. Goetzl, J.-H. Jeoung, S. E. Hennig, H. Dobbek, Structural basis for electron and methyl-group transfer in a methyltransferase system operating in the reductive acetyl-coA pathway. *J. Mol. Biol.* **411**, 96–109 (2011).
18. S. D. Smith *et al.*, Site-directed mutagenesis of HgcA and HgcB reveals amino acid residues important for mercury methylation. *Appl. Environ. Microbiol.* **81**, 3205–3217 (2015).
19. D. H. Haft, M. Gwadz, Eight unexpected selenoprotein families in organometallic biochemistry in *Clostridium difficile*, in ABC transport, and in methylmercury biosynthesis. *J. Bacteriol.* **205**, e00259-22 (2023).
20. D. L. Roberts, S. Zhao, T. Doukov, S. W. Ragsdale, The reductive acetyl coenzyme A pathway: Sequence and heterologous expression of active methyltetrahydrofolate:Corrinoid/iron-sulfur protein methyltransferase from *Clostridium thermoaceticum*. *J. Bacteriol.* **176**, 6127–6130 (1994).
21. T. Doukov, J. Seravalli, J. J. Stezowski, S. W. Ragsdale, Crystal structure of a methyltetrahydrofolate- and corrinoid-dependent methyltransferase. *Structure* **8**, 817–830 (2000).
22. T. I. Doukov, H. Hemmi, C. L. Drennan, S. W. Ragsdale, Structural and kinetic evidence for an extended hydrogen-bonding network in catalysis of methyl group transfer. *J. Biol. Chem.* **282**, 6609–6618 (2007).
23. T. B. Demissie, B. D. Garabato, K. Ruud, P. M. Kozlowski, Mercury methylation by cobalt corrinoids: Relativistic effects dictate the reaction mechanism. *Angew. Chem. Int. Ed. Engl.* **55**, 11503–11506 (2016).
24. J. Zhou, D. Riccardi, A. Beste, J. C. Smith, J. M. Parks, Mercury methylation by HgcA: Theory supports carbanion transfer to Hg(II). *Inorg. Chem.* **53**, 772–777 (2014).
25. S. S. Date *et al.*, Kinetics of enzymatic mercury methylation at nanomolar concentrations catalyzed by HgcAB. *Appl. Environ. Microbiol.* **85**, e00438-19 (2019).
26. C. Qian *et al.*, Quantitative proteomic analysis of biological processes and responses of the bacterium *Desulfovibrio desulfuricans* ND132 upon deletion of its mercury methylation genes. *Proteomics* **18**, 1700479 (2018).
27. C. C. Gilmour *et al.*, *Pseudodesulfovibrio mercurii* sp. nov., a mercury-methylating bacterium isolated from sediment. *Int. J. Syst. Evol. Microbiol.* **71**, p004697 (2021).
28. C. T. Chung, S. L. Niemela, R. H. Miller, One-step preparation of competent *Escherichia coli*: Transformation and storage of bacterial cells in the same solution. *Proc. Natl. Acad. Sci. U.S.A.* **86**, 2172–2175 (1989).
29. M. Nakamura, K. Saeki, Y. Takahashi, Hyperproduction of recombinant ferredoxins in *Escherichia coli* by coexpression of the ORF1-ORF2-iscS-iscU-iscA-hscB-hscA-fdx-ORF3 gene cluster. *J. Biochem. (Tokyo)* **126**, 10–18 (1999).
30. D. C. Johnson, D. R. Dean, A. D. Smith, M. K. Johnson, Structure, function, and formation of biological iron-sulfur clusters. *Annu. Rev. Biochem.* **74**, 247–281 (2005).
31. C.-W. Lin, J. W. McCabe, D. H. Russell, D. P. Barondeau, Molecular mechanism of ISC iron–sulfur cluster biogenesis revealed by high-resolution native mass spectrometry. *J. Am. Chem. Soc.* **142**, 6018–6029 (2020).
32. N. D. Lanz *et al.*, Enhanced solubilization of class B radical *S*-Adenosylmethionine methylases by improved cobalamin uptake in *Escherichia coli*. *Biochemistry* **57**, 1475–1490 (2018).
33. A. Spriestersbach, J. Kubicek, F. Schäfer, H. Block, B. Maertens, "Purification of His-Tagged Proteins" in *Methods in Enzymology*, J. R. Lorsch, Ed. (Elsevier, 2015), pp. 1–15.
34. R. A. Firth *et al.*, The chemistry of vitamin B12. Part IX. Evidence for five-co-ordinate cobalt(III) complexes. *J. Chem. Soc. Inorg. Phys. Theor.* 2419–2428 (1968).
35. C. Giannotti, *B12* (Wiley-Interscience New York, 1982).
36. A. J. Blaszczyk, R. X. Wang, S. J. Booker, "TsrM as a model for purifying and characterizing cobalamin-dependent radical S -adenosylmethionine methylases" in *Methods in Enzymology*, S. S. David, Ed. (Elsevier, 2017), pp. 303–329.
37. P. E. Thomas, D. Ryan, W. Levin, An improved staining procedure for the detection of the peroxidase activity of cytochrome P-450 on sodium dodecyl sulfate polyacrylamide gels. *Anal. Biochem.* **75**, 168–176 (1976).
38. K. Zeamari *et al.*, Tuning the redox properties of a [4Fe-4S] center to modulate the activity of Mo-bisPGD periplasmic nitrate reductase. *Biochim. Biophys. Acta Bioenerg.* **1860**, 402–413 (2019).
39. P. J. H. Daas, W. R. Hagen, J. T. Keltjens, C. van der Drift, G. D. Vogels, Activation mechanism of methanol:5-hydroxybenzimidazolylcobamide methyltransferase from. *J. Biol. Chem.* **271**, 22346–22351 (1996).
40. R. V. Banerjee, R. G. Matthews, Cobalamin-dependent methionine synthase. *FASEB J.* **4**, 1450–1459 (1990).
41. S. R. Harder, W. P. Lu, B. A. Feinberg, S. W. Ragsdale, Spectroelectrochemical studies of the corrinoid/iron-sulfur protein involved in acetyl coenzyme A synthesis by *Clostridium thermoaceticum*. *Biochemistry* **28**, 9080–9087 (1989).
42. R. V. Banerjee, S. R. Harder, S. W. Ragsdale, R. G. Matthews, Mechanism of reductive activation of cobalamin-dependent methionine synthase: An electron paramagnetic resonance spectroelectrochemical study. *Biochemistry* **29**, 1129–1135 (1990).

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43. P. Schrapers *et al.*, Axial ligation and redox changes at the cobalt ion in cobalamin bound to corrinoid iron-sulfur protein (CoFeSP) or in solution characterized by XAS and DFT. *PLoS One* **11**, e0158681 (2016).
44. Z. Li *et al.*, An interprotein Co–S coordination complex in the B_{12}–Trafficking pathway. *J. Am. Chem. Soc.* **142**, 16334–16345 (2020).
45. S. Sato, F. Kudo, T. Eguchi, "Characterization of the cobalamin-dependent radical S-adenosyl-l-methionine enzyme C-methyltransferase Fom3 in fosfomycin biosynthesis" in *Methods in Enzymology*, E. N. G. Marsh, Ed. (Elsevier, 2022), pp. 45–70.
46. H. J. Kim *et al.*, GenK-catalyzed C-6′ Methylation in the biosynthesis of gentamicin: Isolation and characterization of a cobalamin-dependent radical SAM enzyme. *J. Am. Chem. Soc.* **135**, 8093–8096 (2013).
47. H. J. Kim, Y. Liu, R. M. McCarty, H. Liu, Reaction catalyzed by GenK, a cobalamin-dependent radical *S* -adenosyl- L -methionine methyltransferase in the biosynthetic pathway of gentamicin, proceeds with retention of configuration. *J. Am. Chem. Soc.* **139**, 16084–16087 (2017).
48. W. J. Werner *et al.*, In vitro phosphinate methylation by PhpK from *Kitasatospora phosalacinea*. *Biochemistry* **50**, 8986–8988 (2011).
49. H. L. Knox *et al.*, Structural basis for non-radical catalysis by TsrM, a radical SAM methylase. *Nat. Chem. Biol.* **17**, 485–491 (2021).
50. N. D. Lanz, S. J. Booker, Auxiliary iron–sulfur cofactors in radical SAM enzymes. *Biochim. Biophys. Acta* **1853**, 1316–1334 (2015).
51. R. Mathews, S. Charlton, R. H. Sands, G. Palmer, On the nature of the spin coupling between the iron-sulfur clusters in the eight-iron ferredoxins. *J. Biol. Chem.* **249**, 4326–4328 (1974).
52. J. R. Baur, M. C. Graves, B. A. Feinberg, S. W. Ragsdale, Characterization of the recombinant *Clostridium pasteurianum* ferredoxin and comparison of its properties with those of the native protein. *BioFactors Oxf. Engl.* **2**, 197–203 (1990).
53. D. W. Nierenberg *et al.*, Delayed cerebellar disease and death after accidental exposure to dimethylmercury. *N. Engl. J. Med.* **338**, 1672–1676 (1998).
54. S. J. Booker, C. T. Lloyd, Twenty years of radical SAM! The genesis of the superfamily. *ACS Bio. Med. Chem. Au* **2**, 538–547 (2022).
55. A. J. Blaszczyk, S. J. Booker, A (Re)discovery of the Fom3 substrate. *Biochemistry* **57**, 891–892 (2018).
56. B. Wang *et al.*, Stereochemical and mechanistic investigation of the reaction catalyzed by Fom3 from *Streptomyces fradiae*, a cobalamin-dependent radical S- denosylmethionine methylase. *Biochemistry* **57**, 4972–4984 (2018).
57. K. D. Allen, S. C. Wang, Initial characterization of Fom3 from *Streptomyces wedmorensis*: The methyltransferase in fosfomycin biosynthesis. *Arch. Biochem. Biophys.* **543**, 67–73 (2014).
58. H. L. Knox, E. K. Sinner, C. A. Townsend, A. K. Boal, S. J. Booker, Structure of a B12-dependent radical SAM enzyme in carbapenem biosynthesis. *Nature* **602**, 343–348 (2022).
59. M. I. Radle, D. V. Miller, T. N. Laremore, S. J. Booker, Methanogenesis marker protein 10 (Mmp10) from *Methanosarcina acetivorans* is a radical S-adenosylmethionine methylase that unexpectedly requires cobalamin. *J. Biol. Chem.* **294**, 11712–11725 (2019).
60. C. D. Fyfe *et al.*, Crystallographic snapshots of a B12-dependent radical SAM methyltransferase. *Nature* **602**, 336–342 (2022).
61. J. Maillard *et al.*, Characterization of the corrinoid iron-sulfur proteintetrachloroethene reductive dehalogenase of *Dehalobacterrestrictus*. *Appl. Environ. Microbiol.* **69**, 4628–4638 (2003).
62. X. Yin *et al.*, Synergistic effects of a chalkophore, methanobactin, on microbial methylation of mercury. *Appl. Environ. Microbiol.* **86**, e00122-20 (2020).
63. Y. Cai, O. C. Braids, Eds., *Biogeochemistry of Environmentally Important Trace Elements* (American Chemical Society, 2002).

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